Introduction

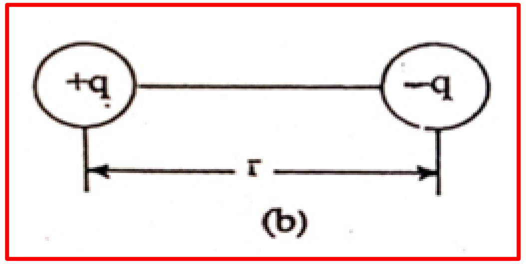
To
Dielectric Properties of Solids

Dielectric It is a non conducting material, such as rubber, glass or wax-paper (insulator).

The energy band model of insulator (dielectric) is like that of a semiconductor; except that the forbidden gap in a dielectric is comparatively large.

Electric dipole

It is formed from two equal and opposite charges with a small distance



The dipole moment of an electric dipole

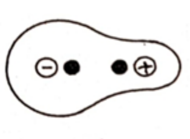
It is a vector from negative to positive charge.

$$\boxed{p = q.r}$$

***In the absence of an external electric field**

In such materials, **electrons are very tightly bound to the atoms**, electrons (do not conduct any electric current).The centers of mass of positive and negative charges are coincide in a molecule of a dielectric, it is ***called non –polar dielectric material.***

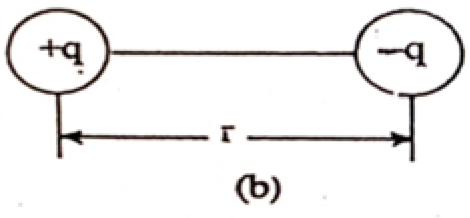
$$\boxed{E_{ex} = o \rightarrow P = 0}$$



****In the presence of an electric field***

the centers of the positive charges are displaced slightly in one direction (direction of field) and the negative charges in the opposite direction. This produces a local electric dipole throughout the sample, it is **called polar dielectric material.**

$$E_{ex} \neq 0 \quad \rightarrow P \neq o$$



For a system, **total electric dipole moment**

$$\boxed{p = \sum_{i=1}^{N} q_i r_i}$$

Polarization The process of electric dipoles alignment, vector.

P = total electric dipole moment/unit Volume

Where N is the number of p per unit volume.

$$\boxed{P = n.p/V = N.p}$$

Dielectric constant (Permittivity of the medium)

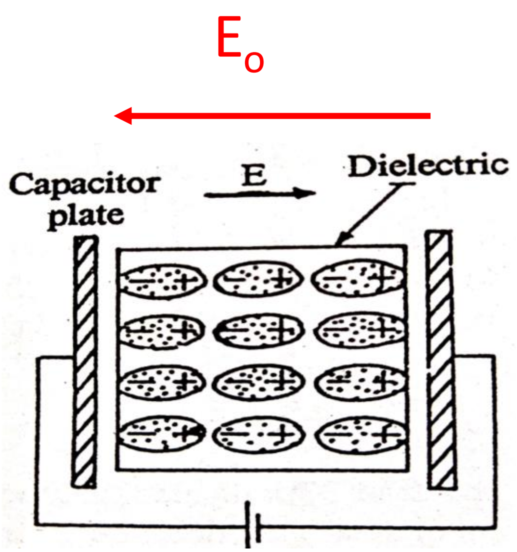
The degree of a medium to resist the flow of charges.

$$\boxed{\epsilon = \epsilon_r \epsilon_o}$$

where ϵ_r is the relative permittivity of the medium and ϵ_o is the permittivity of free space (vacuum)

Dielectric constant and its measurement

*When a dielectric slab is introduced between two parallel plates condenser, the field E_o polarizes the medium, which modifies the field to a new value E.



$\boxed{E_o = V_o/d}$ $\boxed{E = V/d}$ $\boxed{C = Q/V}$

where **d** is the distance between the plates.

Q is the charge on either plate (regardless of sign)

V_o is the potential difference between the plates

For a parallel plate's capacitor, $\boxed{E_o = \sigma/\epsilon_o}$

Where σ is the charge density on each surface. Then $\boxed{\epsilon_r = E_o/E}$ $\boxed{\epsilon_r = V_o/V}$ $\boxed{\epsilon_r = C/C_o > 1}$

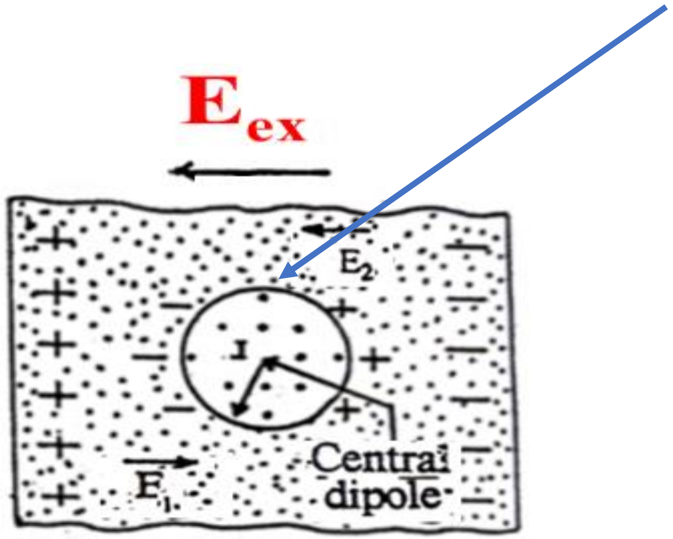
Then the dielectric constant for the medium can be obtained by measuring V_o and V.

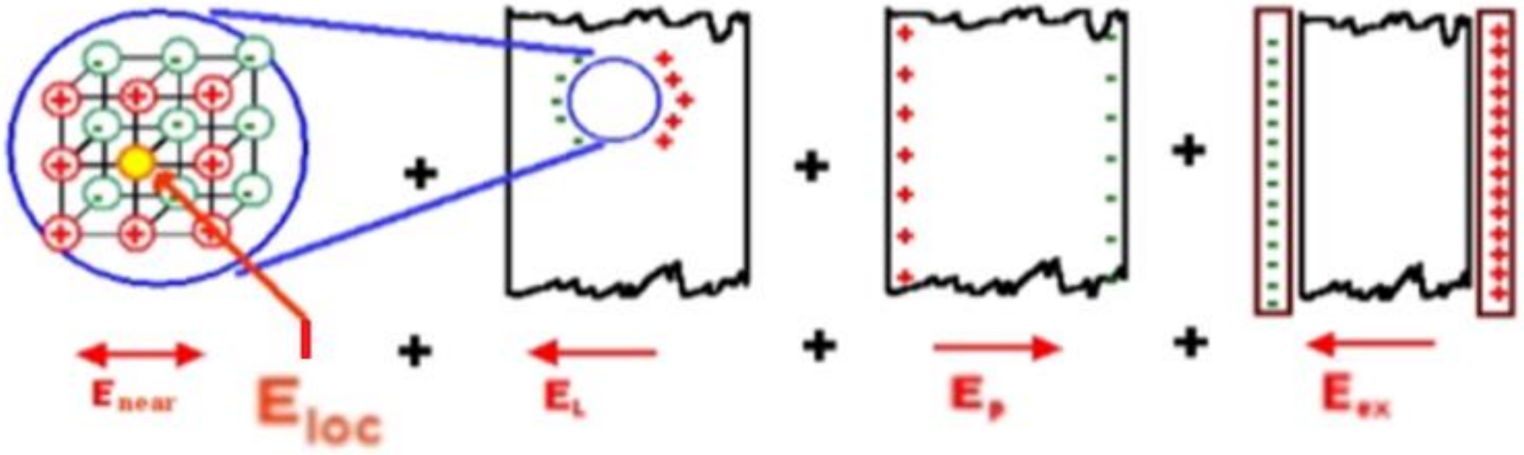
Local Electric field at an atom

Local field is different from the applied external field, E_{ex} (uniform). It acts on a certain reference dipole ***calculated by Lorentz***. He has imagined the reference dipole to be surrounded by ***spherical cavity called Lorentz sphere*** of radius r, sufficiently large as compared to the dimension of the dipole. The space outside the cavity might be treated as a continuous medium as far as the dipole is concerned. **The local field acting on the reference dipole is given by the sum:**

$$\boxed{E_{loc} = E_{ex} + E_{p} + E_{L} + E_{n}}$$

Cavity Material

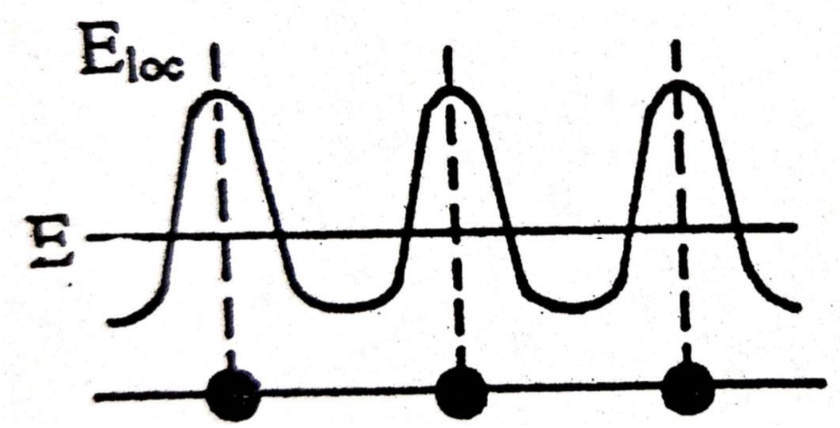




Field at an ion from surrounding | **Lorentz field on the surface** | **Depolarization field of the cavity** | **External field**

$$\boxed{E_{loc} = E\,[(\epsilon_r + 2)/3]}$$ .... *Lorentz equation*

E_{loc} is a microscopic field and is periodic in nature. It is large at molecular sites indicating that the molecules are more effectively polarized than they are under the average field.



Local field ≡ Polarizing field at an atom, molecule or ion in a dielectric, differs from $E_{external}$.